
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, failure analysis, and weight-derived causal structure

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Abstract

1 Can a specialist robot policy be both capable and structurally inspectable? We intro-
2 duce χ -VLA, a 20.1M-parameter vision-language-action policy built entirely from
3 bilinear, linear, and rational learned operations. A mechanical audit finds no soft-
4 max, standard activation, or ordinary normalization, and explicit reconstructions
5 match deployed computation near the float32 precision floor. Trained from scratch
6 on LIBERO-Object, convolutional χ -VLA reaches $93.7\% \pm 0.8\%$ over three seeds
7 under the published 280-step, 50-trial protocol. A three-checkpoint prediction
8 ensemble reaches 96.2% at three times the inference cost. Exact weight-derived
9 attention subspaces outperform equal-rank random controls in all 36 primary
10 block-head tests. At the policy’s visual bond, retaining 64 of 384 causally ranked
11 dimensions preserves 18/20 successes, while a random subspace preserves none.
12 Yet an 80-trial instruction intervention exposes a scene-identity shortcut, and the
13 first direct coefficient-edit screen fails its preregistered selectivity gate. These re-
14 sults establish competitive specialist control and a concrete weight-analysis surface,
15 not zero-shot robotic generalization, robust language grounding, or negligible cost
16 against a matched conventional twin.

17 1 Introduction

18 Most robot policies mix linear maps with softmax attention, smooth activations, and input-dependent
19 normalization. These operations are effective, but they prevent the complete learned map from being
20 represented as one explicit polynomial or rational tensor network. Prior work showed that bilinear
21 language and vision models expose useful weight structure [3, 4]. It remained unclear whether the
22 same restriction could support a continuous closed-loop robot policy.

23 The control setting is demanding for two reasons. First, small action errors compound when the
24 policy observes and corrects its own trajectory. Second, per-instance normalization and multimodal
25 continuous actions do not transfer automatically from the language-model setting. A token model
26 can place a discrete sample outside its learned logit map. A continuous policy must preserve precise
27 actions while avoiding an invalid average of distinct valid modes.

28 We study the narrower but testable question:

29 *Can a specialist VLA retain competitive closed-loop capability when every deployed*
30 *learned operation is polynomial or rational, and does the resulting structure*
31 *provide a useful weight-based analysis surface?*

32 The stronger thesis motivating the next experiments is a three-part conjunction. A compact class of
33 specialist manipulation policies should incur no more than a predeclared capability loss relative to

matched conventional twins. Its trained weights should expose stable behaviorally causal interaction terms. Direct edits to those terms should then produce predicted and selective changes in closed-loop behavior without retraining. Each component has precedent in isolation. Their conjunction in a competitive visuomotor policy is the intended contribution.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace preserves almost all tested behavior, while a random equal-width subspace fails. A separate instruction intervention exposes a real scene-identity shortcut and limits the language-grounding claim.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The three matrices are a CP factorization of the order-three coefficient tensor.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

The deployed function is exactly rational. Projective coordinates carry each activation as a pair (N, D) representing N/D . Linear, bilinear, residual, and normalization operations update the numerator and denominator without introducing a non-rational learned operation.

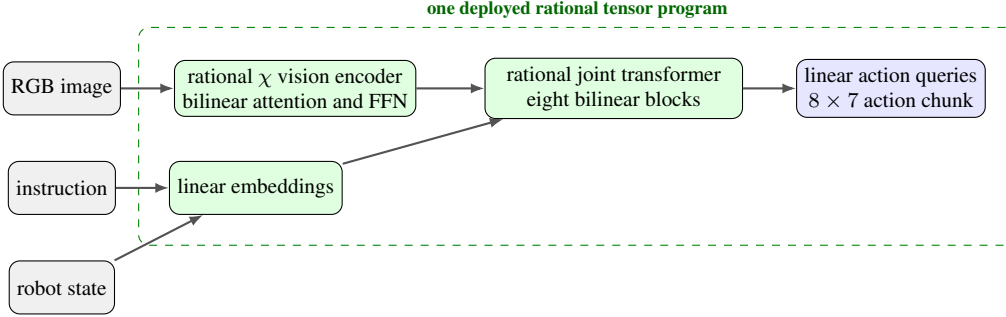


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

69 2.2 Architecture and training

70 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 71 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 72 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 73 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 74 continuous action dimensions. The complete model has 20,137,352 learned parameters.

75 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 76 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 77 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 78 that verify every requested state was loaded.

79 2.3 Mechanical certificate on the trained checkpoint

80 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 81 point at runtime and reconstruct representative operations from saved weights.

82 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 83 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 84 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
 85 summarized in Appendix Figure 7. It does not materialize one compact polynomial for the full
 86 eight-layer transformer.

87 3 Protocol-aligned closed-loop capability

88 3.1 Evaluation protocol

89 We evaluate all ten LIBERO-Object tasks with:

- 90 • 50 canonical trials per task
- 91 • a 280-step cap
- 92 • three independently trained seeds per architecture
- 93 • 500 rollouts per seed and 1,500 rollouts per architecture

94 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 95 rerun in our codebase, so differences may still include implementation and training choices.

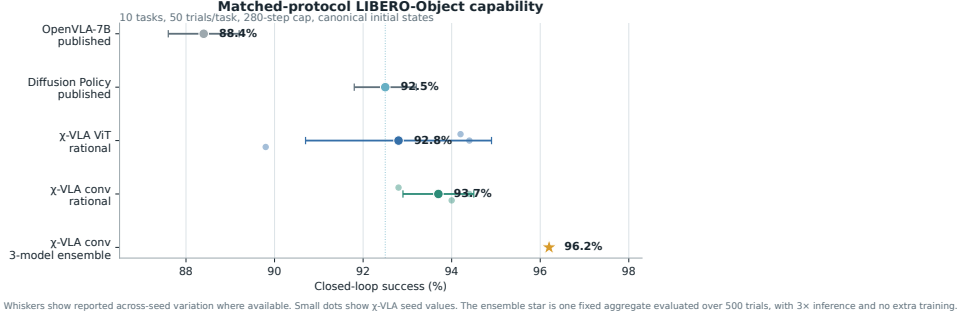


Figure 2: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It does not isolate the cost of decomposability. A decisive cost experiment must train a same-skeleton conventional transformer and staged partial-conversion controls using identical data, updates, seeds, and evaluation states. We treat that experiment as required before restoring the stronger title claim that convertibility is nearly free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient object determined entirely by the weights, mask, and fixed scaling. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$

120 10^{-16} . Agreement with the real trained module ranges from 3.10×10^{-7} to 1.03×10^{-5} because
 121 the deployed implementation intentionally casts the statistic to float32.

122 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 123 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. At toy multi-head scale, the
 124 reduced calculation matches brute-force materialization to 2.13×10^{-14} .

125 4.2 Trained policy result

126 We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final
 127 backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all
 128 evaluation examples. Real cached activations are used only to evaluate how faithfully the selected
 129 weight-derived subspaces preserve the layer’s computation.

130 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 131 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 132 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 133 0, 6, and 7.

134 This extends exact weight-only construction through individual attention layers. It is not an exact
 135 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 136 rapid degree and representation growth.

137 5 Causal policy structure and a language shortcut

138 We next test whether the identified structure affects closed-loop behavior and whether high task
 139 success reflects robust use of the language input.

140 5.1 Causal visual subspace

141 At the visual bond before the joint backbone, we rank directions separately for translation, rotation,
 142 and gripper actions. We compare the selected subspace with random equal-width subspaces in
 143 offline action reconstruction and closed-loop intervention. This ranking uses held-out activations and
 144 downstream sensitivity. It is causal, but not a weight-only discovery result.

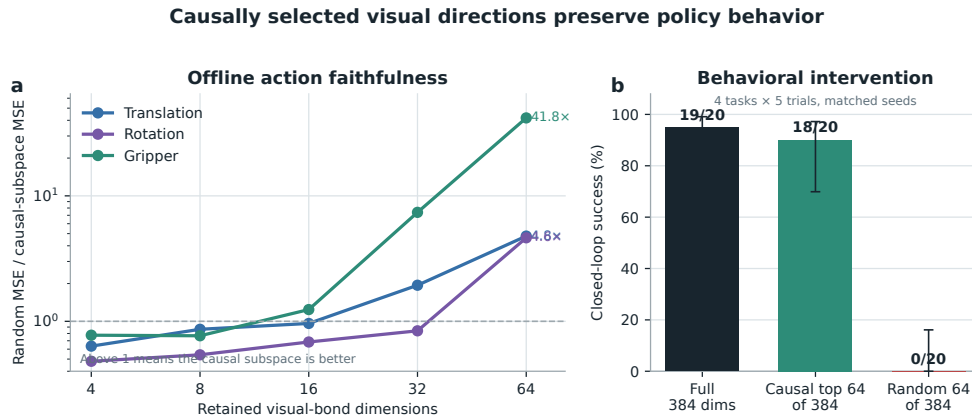


Figure 3: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

145 With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative
 146 to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. In closed loop, the
 147 full representation succeeds in 19/20 trials, the selected subspace in 18/20, and a random subspace
 148 in 0/20.

149 This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and
150 twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream
151 sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same
152 full-suite closed-loop intervention.

153 5.2 Counterfactual instruction test

154 High success does not guarantee robust language use. We hold the image and robot state fixed
155 and replace the instruction with one naming another visible object. Across 80 canonical one-step
156 trials, the predicted reach direction shifts toward the newly named object in only 15.0%. Mean
157 directional cosine to the named object falls from 0.525 under the true instruction to 0.299 under the
158 counterfactual instruction, while cosine to the original target remains 0.464. In an earlier 40-step
159 closed-loop screen, 33.3% finish closer to the new object and 55.6% move measurably toward it.

160 The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target,
161 so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations
162 improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-
163 loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy
164 range.

165 This negative result narrows the VLA claim. The current model is a capable language-conditioned
166 specialist, but it is not yet a robust instruction-grounded policy. LIBERO-CF independently documents
167 the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the
168 shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively
169 repair that shortcut.

170 6 Related work and limitations

171 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
172 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
173 weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained
174 VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a
175 policy designed as a rational tensor program. Tensor and polynomial networks provide the broader
176 multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success
177 can coexist with weak language use [9–11].

178 **Limitations and conclusion.** No same-skeleton conventional twin yet measures conversion cost.
179 The benchmark is one specialist suite, the policy fails a counterfactual instruction test, and exact
180 decomposition is layerwise rather than end to end. The closed-loop subspace test uses one checkpoint
181 and twenty trials, while the first direct coefficient edit fails its selectivity gate. The defensible result
182 is therefore narrow: rational restrictions do not prevent competitive LIBERO-Object control, and
183 they provide a precise weight-derived analysis surface. Broader grounding, matched controls, and
184 validated coefficient edits remain necessary.

185 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
186 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
187 unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

219 A.2 Per-task ensemble behavior

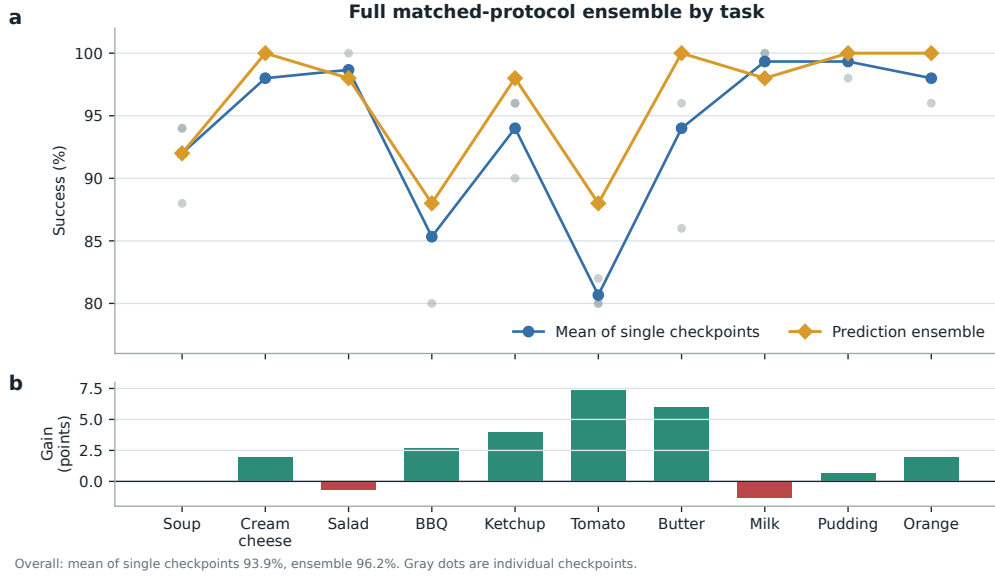


Figure 4: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

220 A.3 Product-routing multimodal head

221 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

222 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 223 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 224 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 225 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

226 A.4 Development interventions

227 Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 228 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 229 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 230 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 231 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 232 points. These are development diagnostics, not validated paper contributions.

What the July 30 intervention experiments taught us

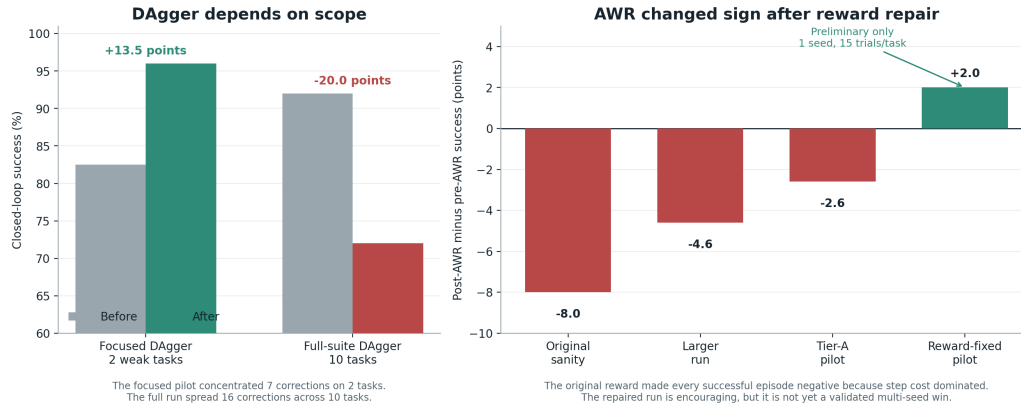


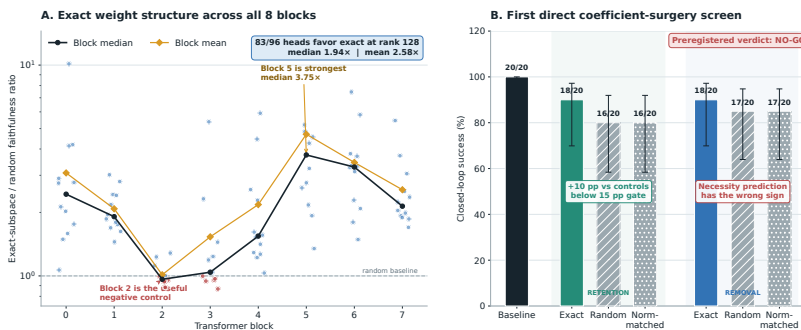
Figure 5: **Development interventions reveal scope and reward sensitivity.** Focused DAGger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

233 A.5 Expanded attention screen and first direct edit

234 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 235 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 236 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 237 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 238 the wider tested rank.

239 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 240 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 241 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 242 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 243 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 244 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A: 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B: 4 tasks × 5 paired canonical episodes. Direct QKV weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 6: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

245 A.6 Numerical scope of rational conversion

246 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 247 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 248 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 249 establish practical numerical safety under distribution shift.

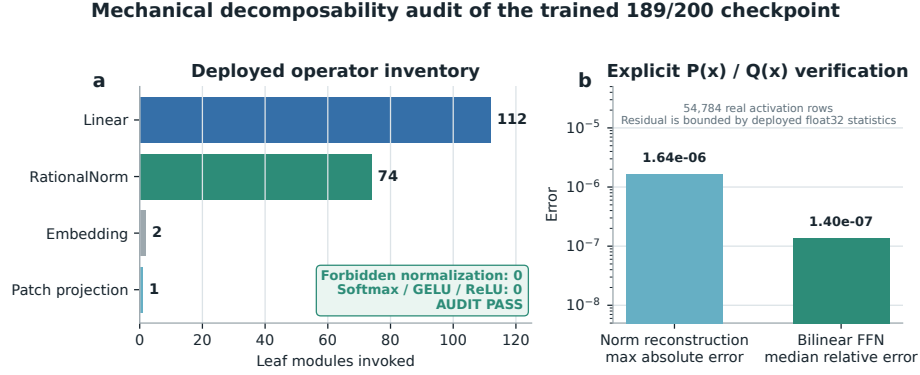


Figure 7: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.